Machine Learning

Assignment 2 Slide Deck

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Datasets

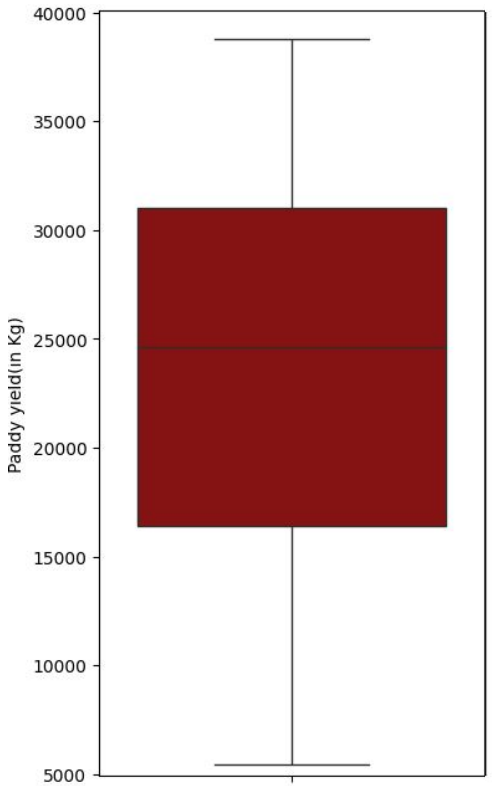
Selection Criteria

- Regression task
- Published after 2020
- Sample size variety
- Feature dimension variety

	Rice Paddy Yield	Utrecht Housing	Student Performance
Year Published	2025	2024	2024
Sample Size	2789	100	10000
Number of Features	44	14	5

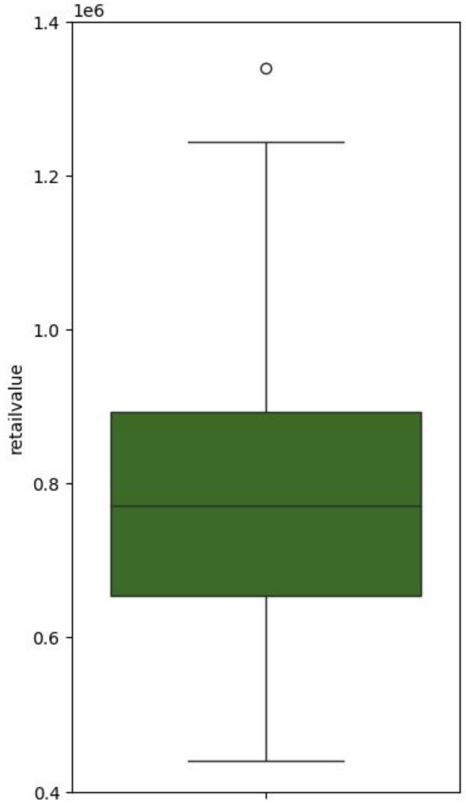
Dataset 1

- [Rice Paddy Yield]
- **2789** Samples
- **44** Features
- Target variable: **Paddy yield(in Kg)**
- Geographical, agricultural, and environmental conditions that affect rice farming output.
- The target variable is the rice output per paddy.
- The target variable is multimodal.



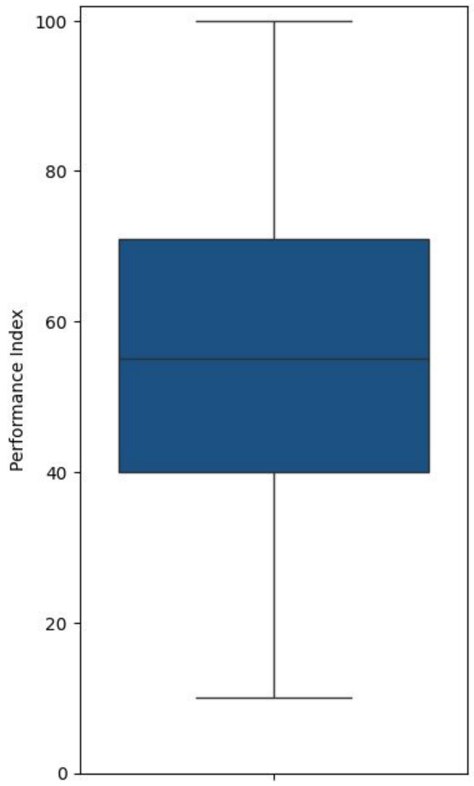
Dataset 2

- [Utrecht Housing]
- **100** Samples
- **14** Features
- Target variable: **retailvalue**
- Details on the plot and architecture of houses in Utrecht.
- The target variable is a house's asking price.
- The target variable is normally distributed.



Dataset 3

- Student Performance
- **10000** Samples
- **5** Features
- Target variable: **Performance Index**
- Factors influencing academic student performance.
- The target variable is a student's final grade.
- The target variable is normally distributed.



Algorithms

Regression Tree

- Model built by **recursively partitioning** dataset into subsets using input features.
- Recursive splitting **repeats until stopping criteria** is met:
    - Maximum tree depth
    - Minimum number of samples to split
    - Minimum number of samples in leaf
    - No split reduces error
- Each tree **level** introduces a **new partitioning criteria**.
- The **internal nodes** represent a **decision on a feature**.
- The **leaf nodes** represent a **numeric prediction** (mean of data points in node).
- Select split that **minimizes sum of squared errors**.

Regression Tree

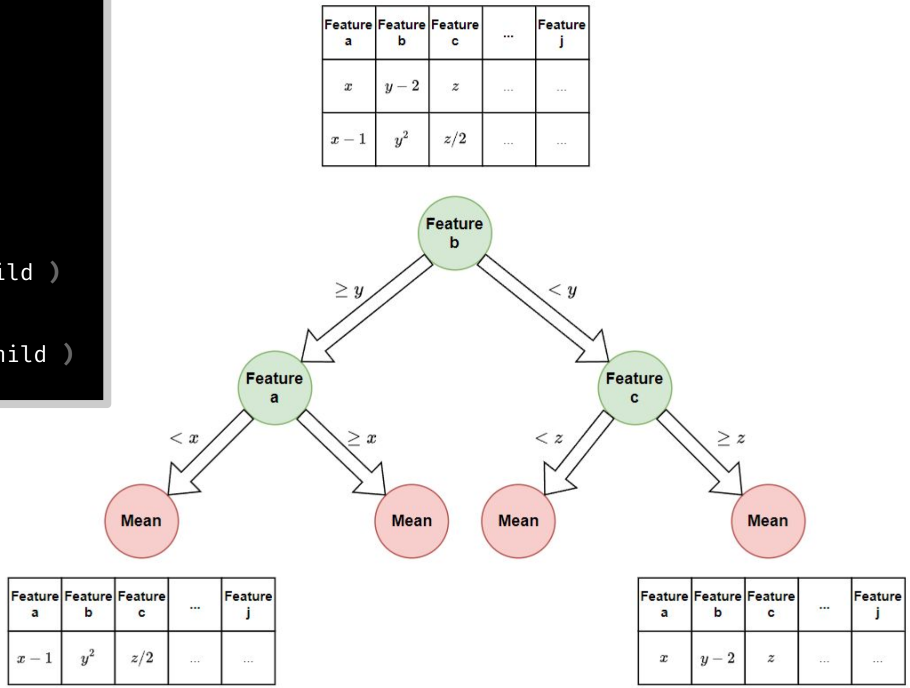
Pseudocode to Build Tree from Data

```
function Recursively_Build_Regression_Tree ( data, current_depth )

    node = Build_Node ( )
    node.prediction = Calculate_Mean ( data )

    if current_depth > maximum_depth :
        Make_Leaf ( node )

    if node_samples < minimum_samples :
        Make_Leaf ( node )

    for feature f in available_features :
        for partition p in possible_splits :

            left_of_p, right_of_p = Divide_Data_On_Feature ( f, p )
            error = Calculate_Error ( left_of_p ) + Calculate_Error ( right_of_p )

            if error < best_error:
                best_error = error
                Save_Split ( left_of_p, right_of_p )

    if not Found_Improving_Split ( ) :
        Make_Leaf ( node )
    else :
        node.left_child = Recursively_Build_Tree ( left_of_p, current_depth + 1 )
        node.right_child = Recursively_Build_Tree ( right_of_p, current_depth + 1 )

    return node
```

Regression Tree

Pseudocode to Predict Datapoint

```
function Recursively_Predict_From_Tree ( datapoint, node )

    if node.children == 0 :
        return node.prediction # Mean

    # Evaluate datapoint using
    f = node.feature
    value = datapoint [ f ]
    p = node.partition

    if value < p :
        return Recursively_Predict ( datapoint, node.left_child )

    else :
        return Recursively_Predict ( datapoint, node.right_child )
```



Regression Tree

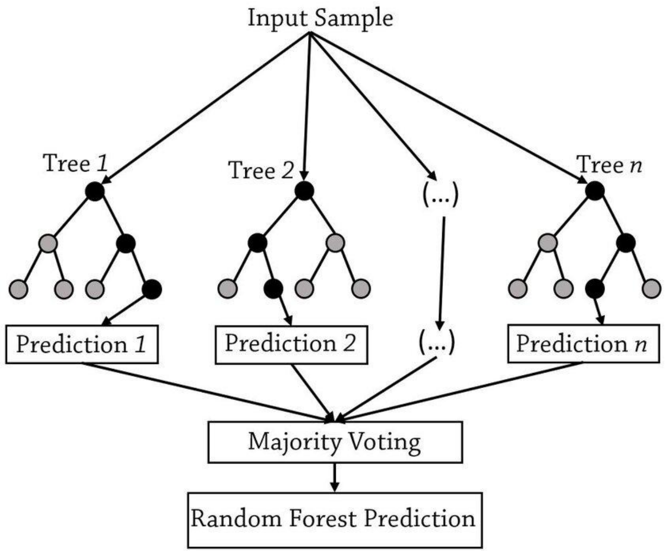
Challenges

- **Multiple possible** splitting and stopping **criteria**.
  - Only a sunset were implemented
- Managing **recursion depth** and computational complexity.
- Efficiency concerns on the naive implementation.
- Significant **performance gains** by leveraging the **sum of square errors**.

$$\sum(y-\hat{y})^2 = \sum(y^2) - \frac{\sum(y)^2}{n}$$

Random Regression Forest

- Ensemble of multiple regression trees.
- For each tree is **trained on a bootstrapped subset** of the data.
- For each tree a random subset of features is selected to be considered.
  - Can be limited with a parameter
- Final prediction is the **average of all** different regression **tree estimates**.



Random Regression Forest

Pseudocode

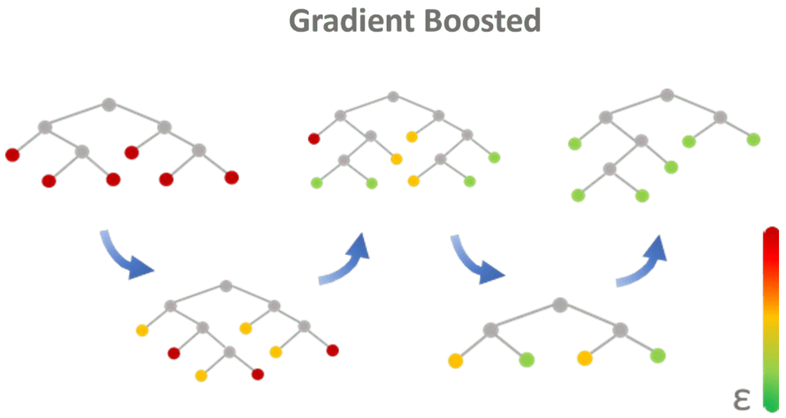
Build Ensemble Model

```
function Iteratively_Build_Random_Forest ( data, forest_size )

    forest = [ ]

    while Length_Of ( forest ) < forest_size :

        bootstrap_data = Sample_With_Replacement ( data )
        bootstrap_data.features = Select_Random_Subset ( data.features )

        tree = Recursively_Build_Tree ( bootstrap_data )
        Append ( forest, tree )

    return forest
```

Predict Datapoint

```
function Iteratively_Predict_From_Forest ( datapoint, forest )

    predictions = [ ]

    for tree in forest :

        # Select only relevant
        datapoint.features = tree.required_features

        prediction = Recursively_Predict ( datapoint, tree )
        Append ( predictions, prediction )

    return Calculate_Mean ( predictions )
```

Boosted Regression Forest



- Each tree is trained to **correct the errors** of the ensemble so far.
- The bootstrap sample is taken with **higher likelihood** of selecting the data points that contribute the **most error**.

Boosted Regression Forest

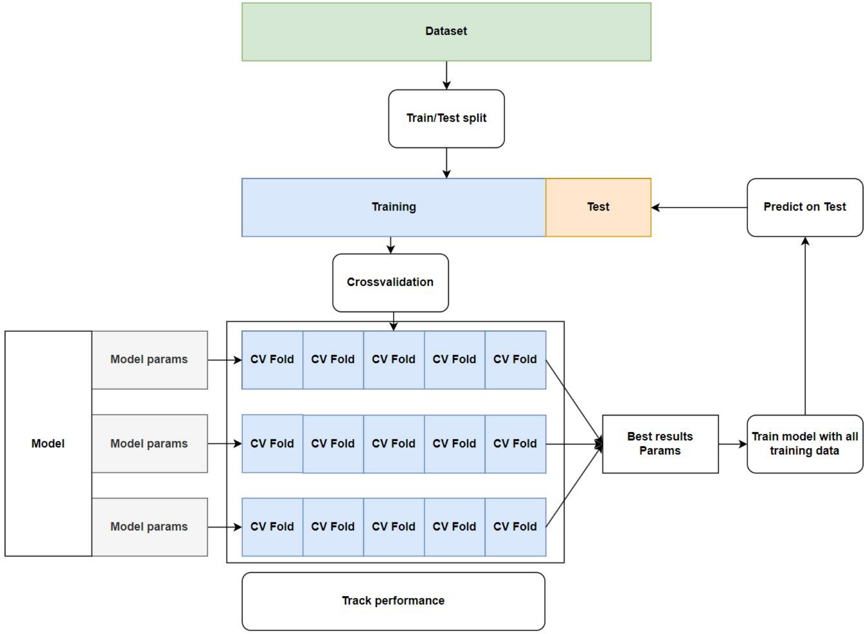
Pseudocode to Build Ensemble Model

```
function Iteratively_Build_Boosted_Forest ( data, forest_size )

    forest = [ ]
    probability = Initialize_With_Uniform_Probability ( )

    while Length_Of ( forest ) < forest_size :

        bootstrap_data = Sample_With_Replacement ( data, probability )
        bootstrap_data.features = Select_Random_Subset ( data.features )

        tree = Recursively_Build_Tree ( bootstrap_data )
        Append ( forest, tree )

        prediction = Predict_All ( forest, data )
        residuals = Calculate_Error ( data.target, prediction )
        probability = Update_From ( residuals )

    return forest
```

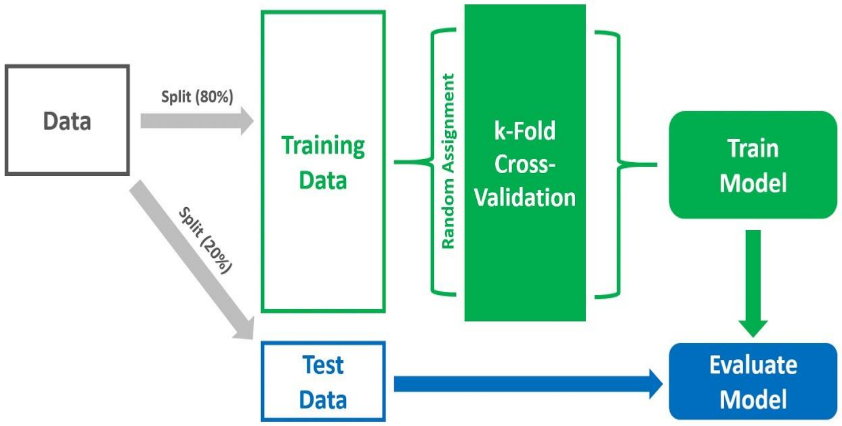
Experiments

Experiment design

1. The dataset was divided into **Training** and **Test** sets.
2. The **Training** set was further split into multiple **Validation** sets for cross-validation.
3. Various parameter combinations were evaluated during cross-validation.
4. The final model, used to measure holdout (**Test** set) performance, was trained on the entire Training data using the best-performing parameter combination.



Cross-validation



- K-fold cross-validation was used.
- Each data point is included in both training and validation sets across different folds.
- This approach assesses model performance under various sample distributions.
- Evaluation metrics are averaged over all folds.
- The variability (spread) of the metrics across folds is also measured.

Metrics

RMSE

- Measured in original unit, high interpretability
    - Cannot be compared across datasets
- Larger errors are penalized harder
- Sensitive to outliers

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

MAPE

- Measured in percentage, scale independent, high interpretability
    - Can be compared across datasets
- Less sensitive to outliers
- Unstable when true or predicted values close to 0

$$MAPE = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

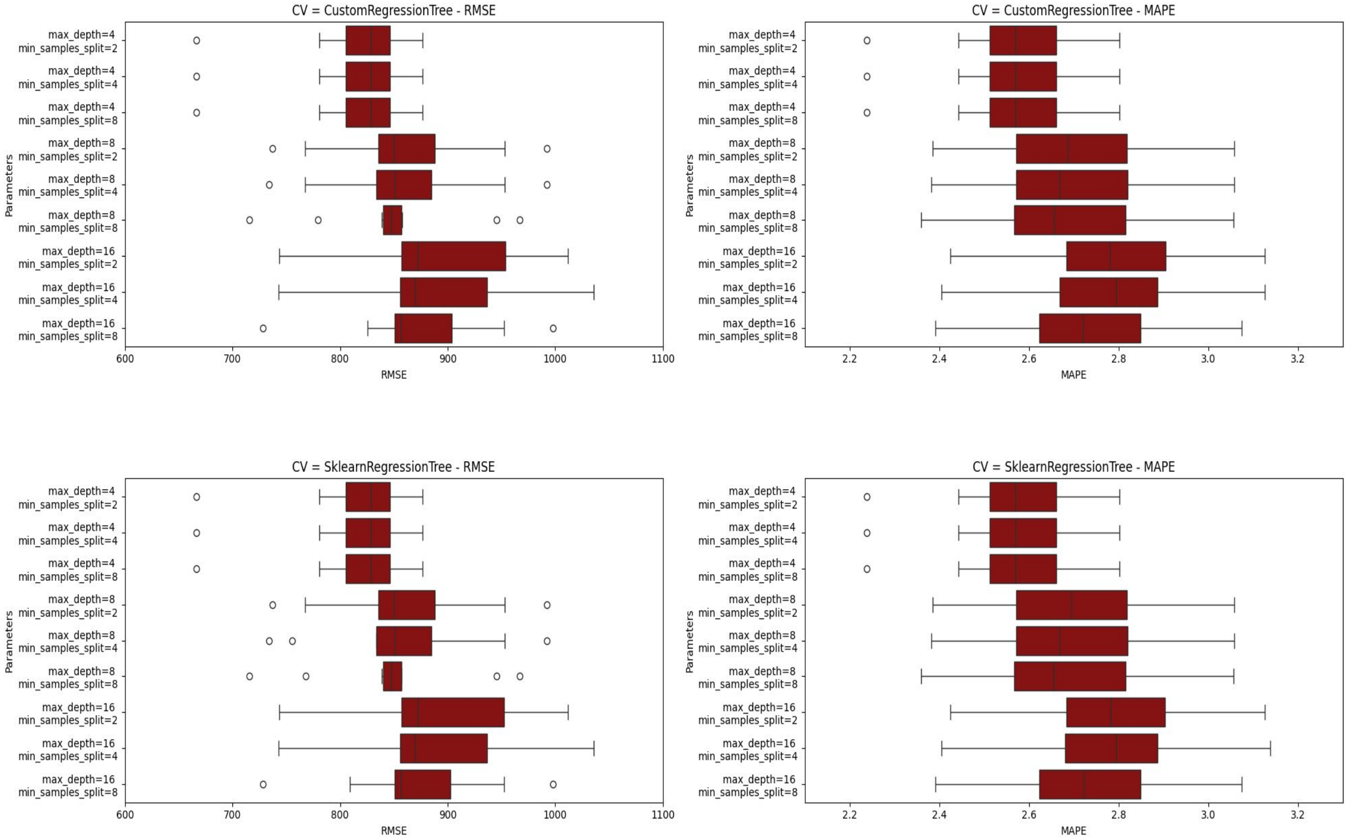
Results

Cross-validation Results
Dataset 1

Custom Regression Tree

vs

Sklearn Regression Tree

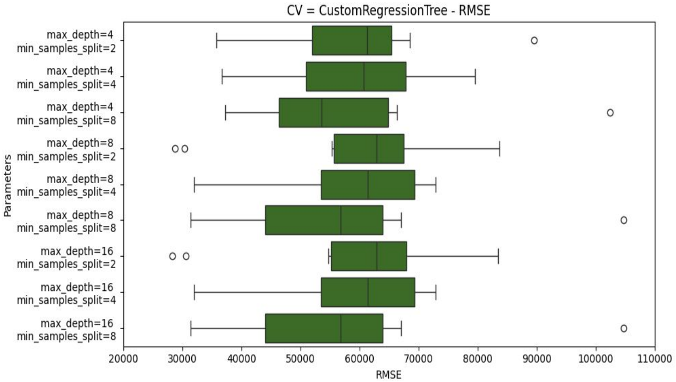


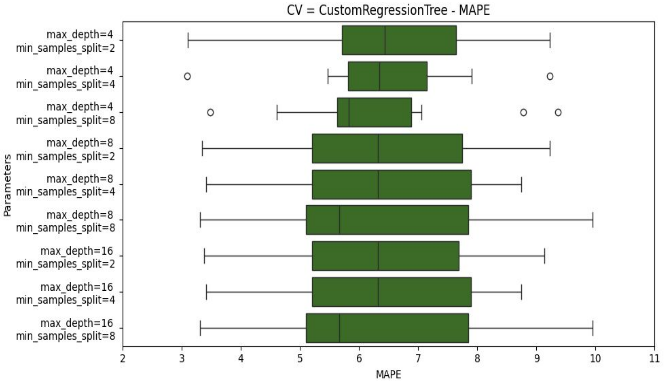
Cross-validation Results
Dataset 2

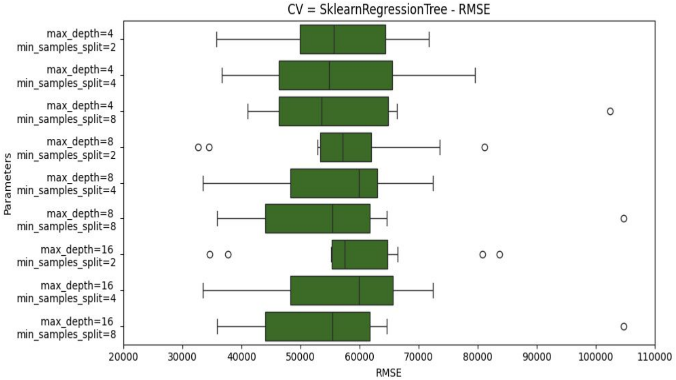
Custom Regression Tree

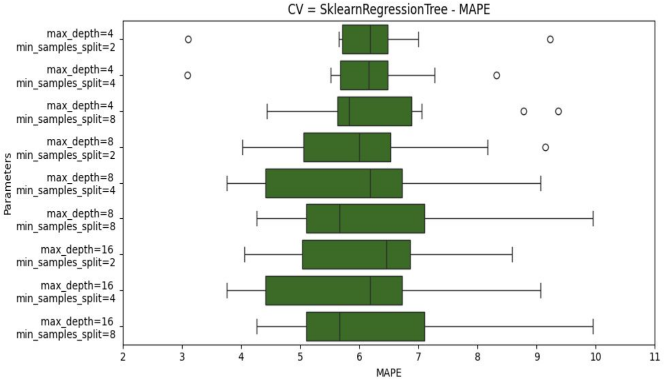
vs

Sklearn Regression Tree







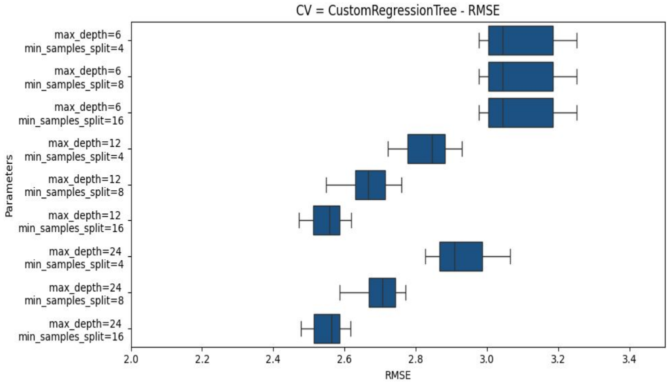


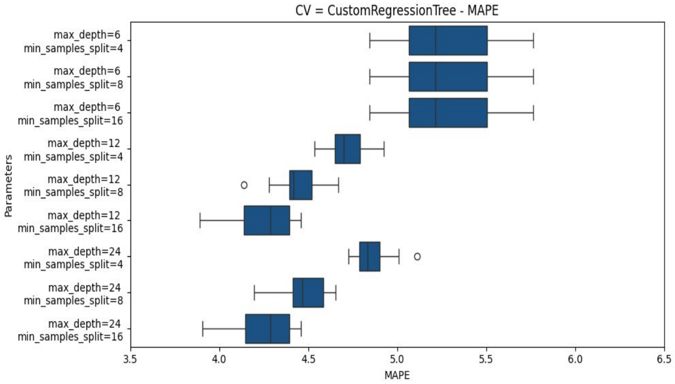
Cross-validation Results
Dataset 3

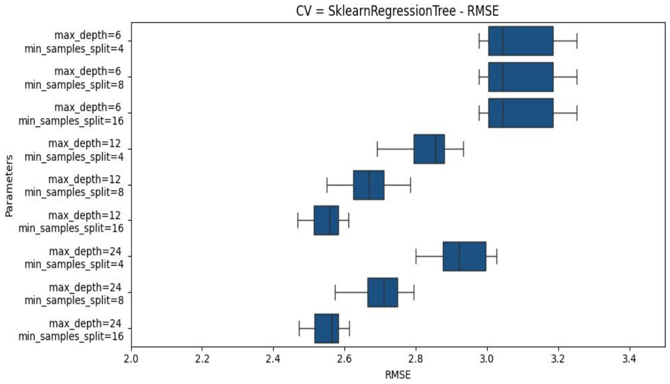
Custom Regression Tree

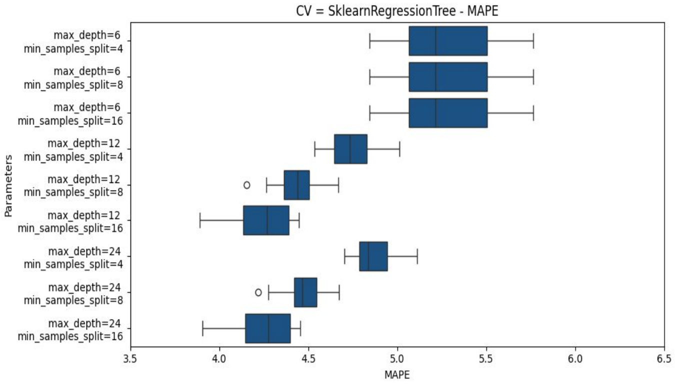
vs

Sklearn Regression Tree









Cross-validation Results
Dataset 1

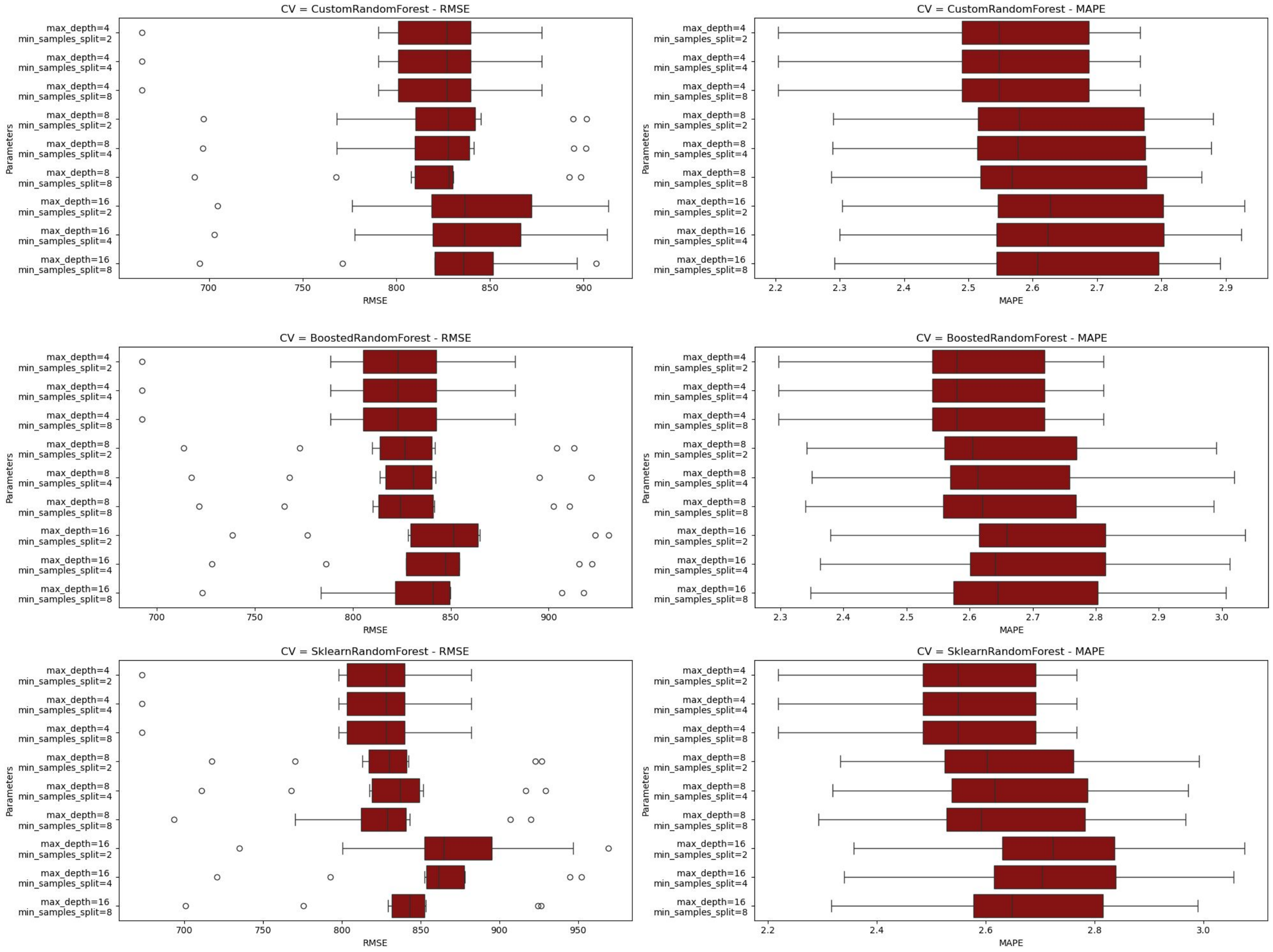
Custom Random Forest

vs

Custom Boosted Forest

vs

Sklearn Random Forest



Cross-validation Results
Dataset 2

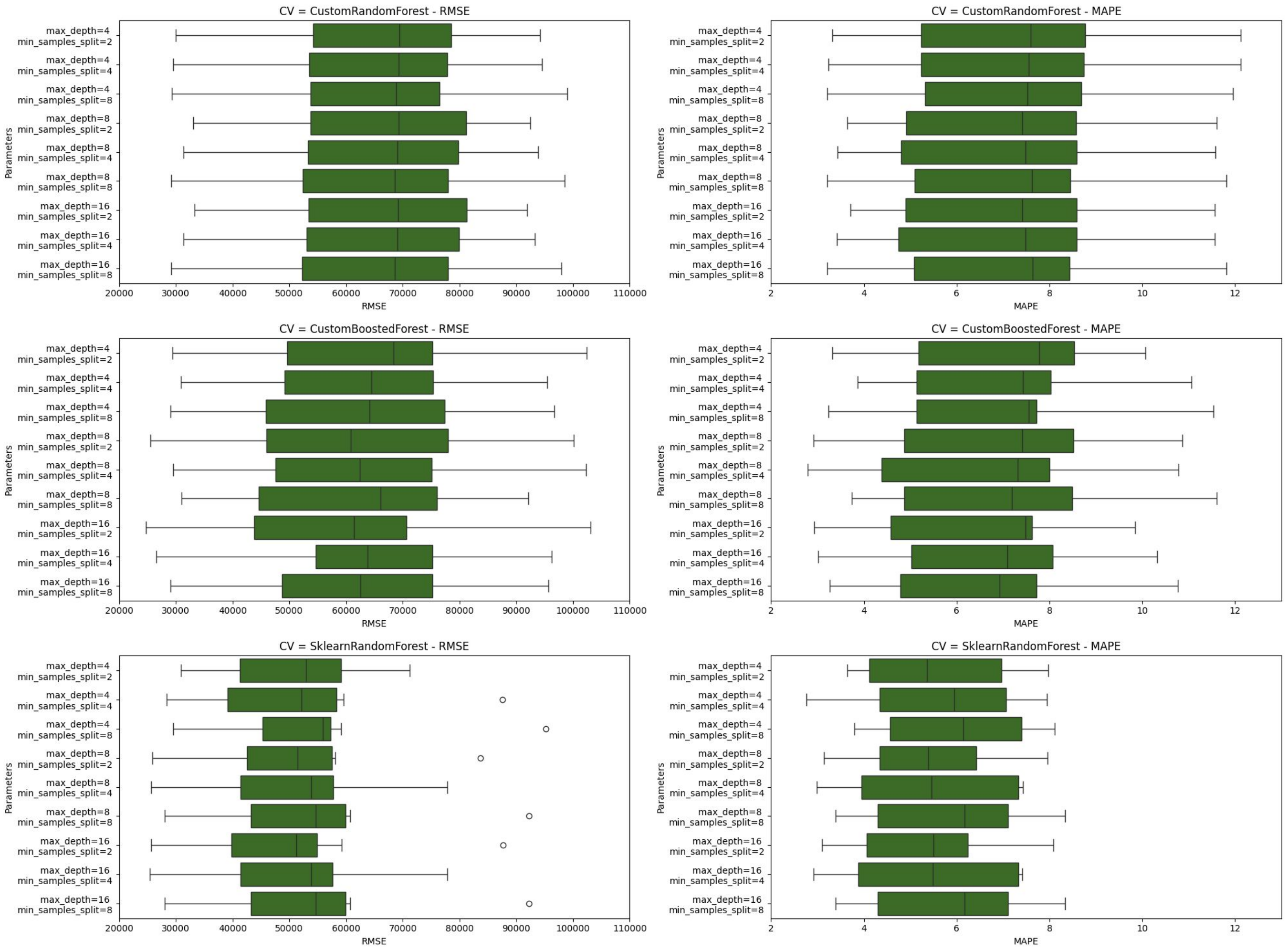
Custom Random Forest

vs

Custom Boosted Forest

vs

Sklearn Random Forest



Cross-validation Results
Dataset 3

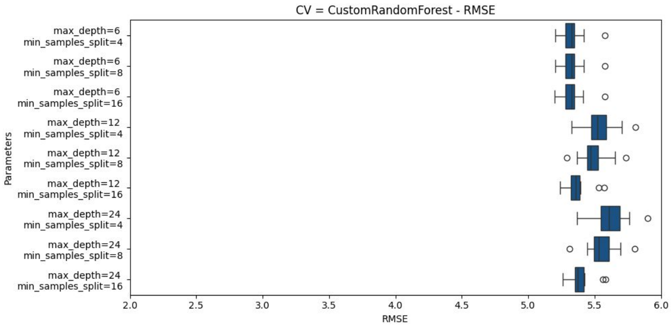
Custom Random Forest

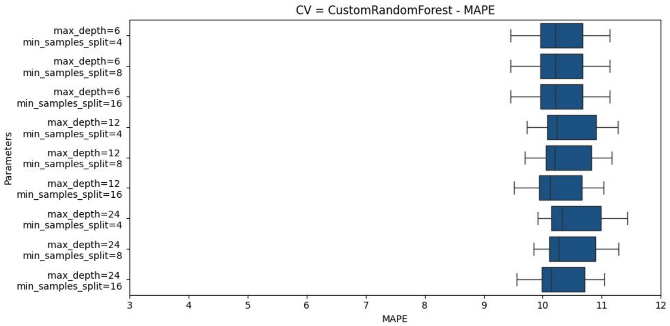
vs

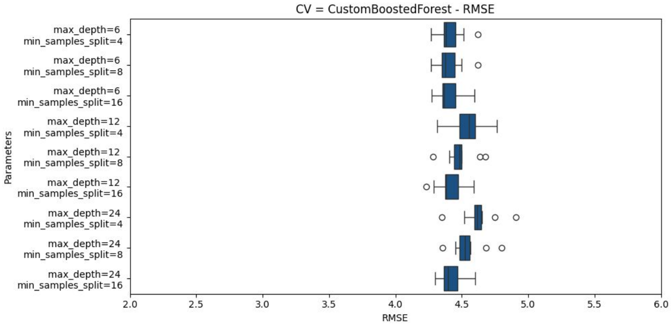
Custom Boosted Forest

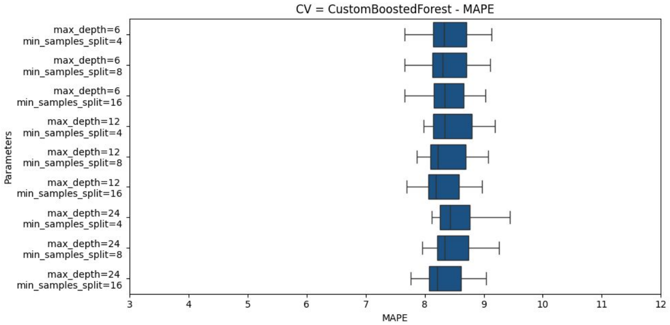
vs

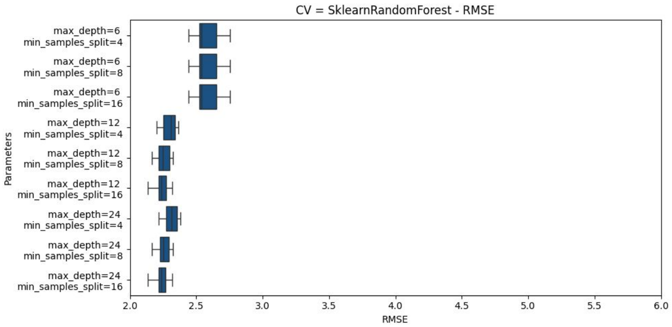
Sklearn Random Forest

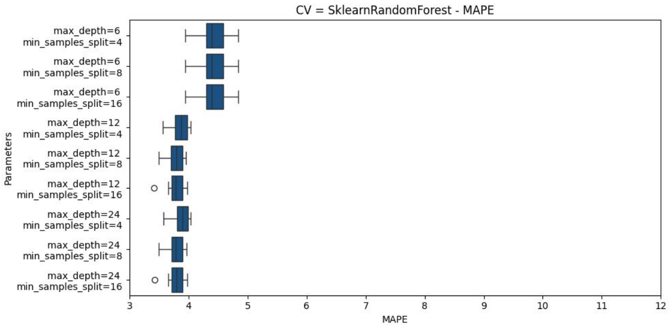




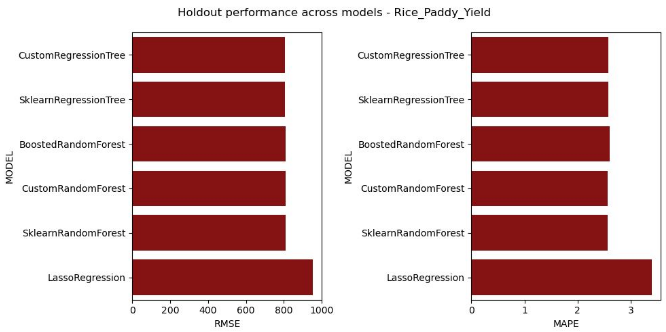


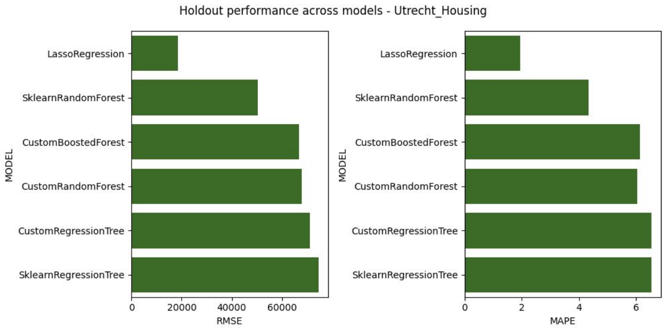


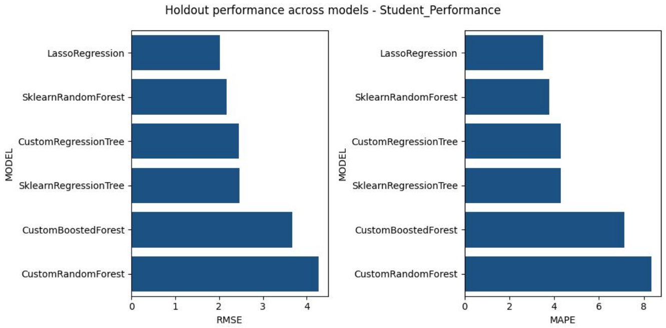




	Model	Max Depth	Minimum Samples Split	Alpha	RMSE	MAPE
Rice Paddy Yield	Custom Regression Tree	4	2		**806.22**	**2.58**
	Scikit-learn Regression Tree	4	2		**806.22**	**2.58**
	Custom Random Forest	16	2		2605.13	15.72
	Custom Boosted Forest	4	2		5586.73	33.90
	Scikit-learn Random Forest	8	2		835.16	2.68
	Scikit-learn Lasso Regression			10	952.49	3.39
Utrecht Housing	Custom Regression Tree	4	4		71029.79	6.54
	Scikit-learn Regression Tree	4	2		74502.53	6.54
	Custom Random Forest	16	2		67904.81	6.05
	Custom Boosted Forest	16	8		66902.46	6.13
	Scikit-learn Random Forest	4	2		50278.56	4.34
	Scikit-learn Lasso Regression			1000	**18666.57**	**1.96**
Student Performance	Custom Regression Tree	12	16		2.46	4.28
	Scikit-learn Regression Tree	12	16		2.46	4.29
	Custom Random Forest	6	16		4.27	8.35
	Custom Boosted Forest	6	16		3.67	7.14
	Scikit-learn Random Forest	24	16		2.18	3.79
	Scikit-learn Lasso Regression			0.01	**2.02**	**3.51**



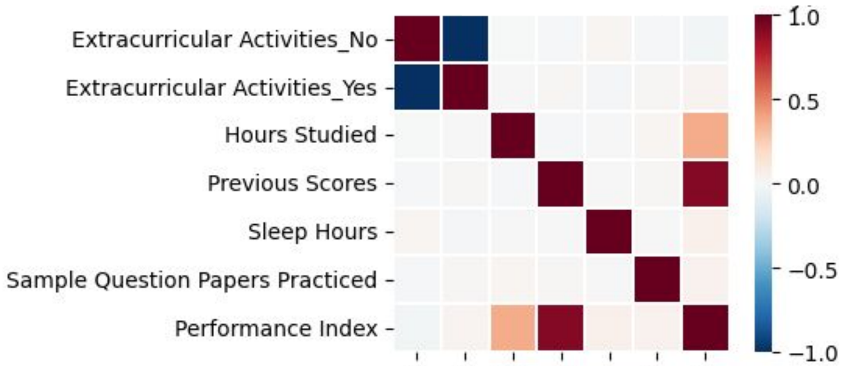




Conclusions

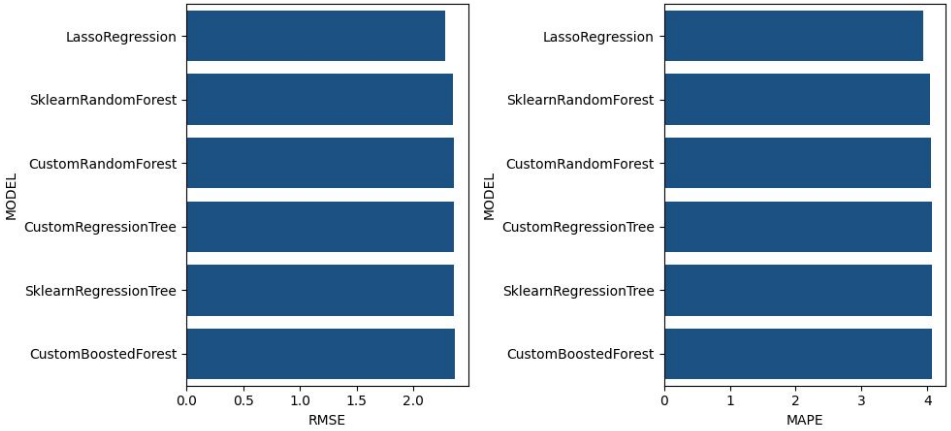
Conclusions

- Why single tree outperforms ensemble methods on **datasets 3**?
  - Boosted forest has worst performance hints at…
  - Ensemble models are likely amplifying noise.



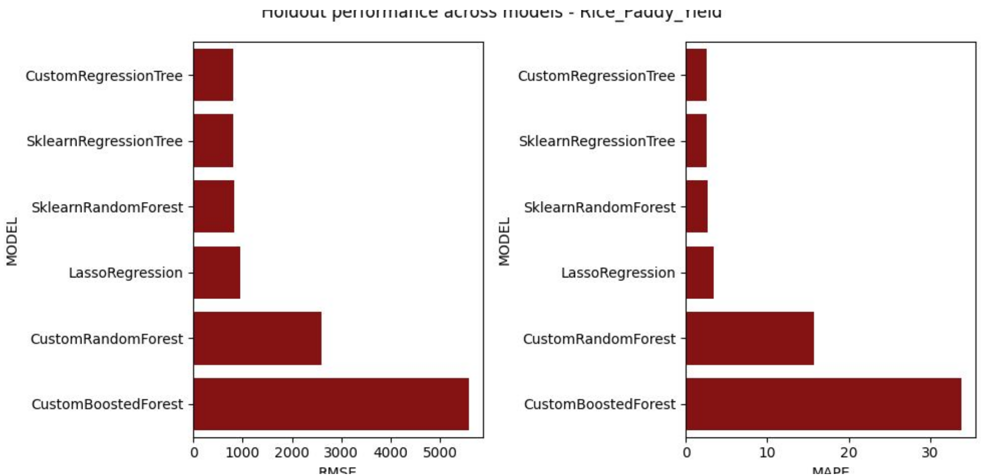
Conclusions

- Rerun variation of experiment to explore issue further.
    - Only use 2 highly correlated features for dataset 3.
- **Result:** Ensemble methods now perform similarly to all other models.



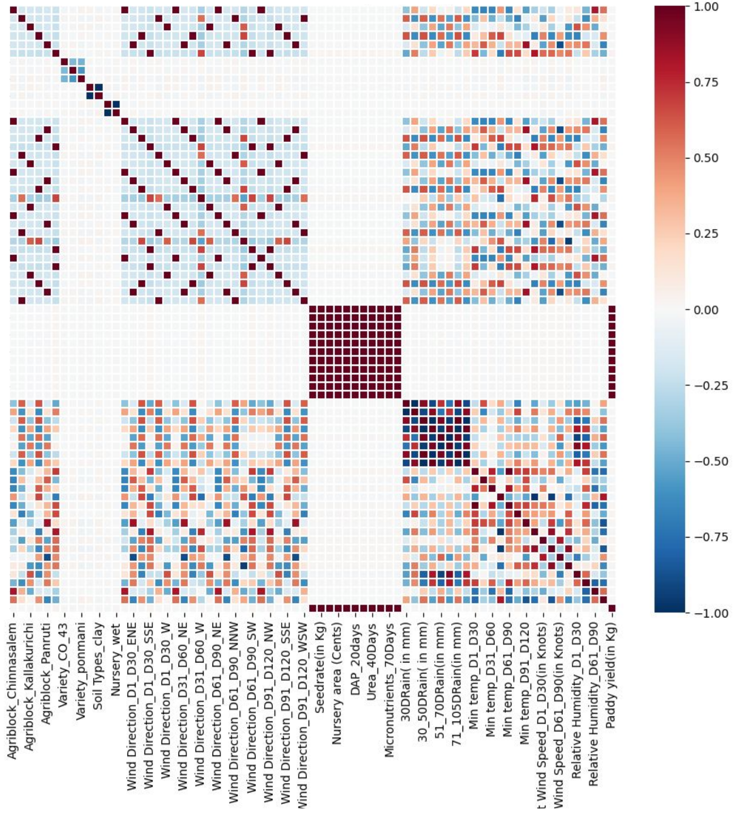
Conclusions

- Conversely, if we reduce the number of **MAX_FEATURES** ensemble methods can sample to generate a tree on **datasets 1** we can artificially induce the noise amplification and get ....
    - Boosted forest again with worst performance.



Conclusions

- This is due to the generated trees rarely getting access to one of the only eleven meaningful features.
  - Originally **MAX_FEATURES** was set to the available number of features (**44**).
  - Rerun limited **MAX_FEATURES** (to **5**).
- **Result:** Custom ensemble methods now perform poorly.



Appendix

Cross-validation Results
Dataset 1

Custom Regression Tree

vs

Sklearn Regression Tree

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=4 min_samples_split=2	2.566070	813.030728	2.237085	666.576743	2.801338	876.750766	0.159264	58.276346	2.725335	871.307074
max_depth=4 min_samples_split=4	2.566070	813.030728	2.237085	666.576743	2.801338	876.750766	0.159264	58.276346	2.725335	871.307074
max_depth=4 min_samples_split=8	2.566070	813.030728	2.237085	666.576743	2.801338	876.750766	0.159264	58.276346	2.725335	871.307074
max_depth=8 min_samples_split=8	2.686063	849.918399	2.359543	715.980326	3.055969	967.066376	0.210451	71.507999	2.896515	921.426398
max_depth=8 min_samples_split=4	2.695760	857.697288	2.382044	734.038391	3.057102	992.200492	0.205679	76.750343	2.901439	934.447631
max_depth=8 min_samples_split=2	2.699423	858.731244	2.384156	736.840483	3.057102	992.200492	0.205013	76.305970	2.904436	935.037214
max_depth=16 min_samples_split=8	2.739814	870.575454	2.390086	728.038295	3.073541	997.702798	0.200421	73.275344	2.940235	943.850798
max_depth=16 min_samples_split=2	2.793034	888.302701	2.423666	743.262360	3.125381	1011.905178	0.213002	79.371993	3.006036	967.674694
max_depth=16 min_samples_split=4	2.785385	888.390779	2.404627	743.065415	3.125335	1036.035906	0.218638	81.849824	3.004023	970.240603

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=4 min_samples_split=2	2.566070	813.030728	2.237085	666.576743	2.801338	876.750766	0.159264	58.276346	2.725335	871.307074
max_depth=4 min_samples_split=4	2.566070	813.030728	2.237085	666.576743	2.801338	876.750766	0.159264	58.276346	2.725335	871.307074
max_depth=4 min_samples_split=8	2.566070	813.030728	2.237085	666.576743	2.801338	876.750766	0.159264	58.276346	2.725335	871.307074
max_depth=8 min_samples_split=8	2.684964	848.635535	2.359543	715.980326	3.055969	967.066376	0.211307	72.821183	2.896271	921.456718
max_depth=8 min_samples_split=4	2.695016	856.537120	2.382044	734.038391	3.057102	992.200492	0.206495	78.339227	2.901511	934.876348
max_depth=8 min_samples_split=2	2.701056	858.741926	2.384156	736.840483	3.057102	992.200492	0.204609	76.302970	2.905665	935.044896
max_depth=16 min_samples_split=8	2.738923	868.825498	2.390086	728.038295	3.073541	997.702798	0.201352	74.459861	2.940275	943.285359
max_depth=16 min_samples_split=2	2.790526	884.202989	2.423666	743.262360	3.125381	1011.936801	0.217319	83.577340	3.007845	967.780329
max_depth=16 min_samples_split=4	2.789650	888.424109	2.404627	743.065415	3.138015	1036.066792	0.220756	81.852881	3.010406	970.276990

Cross-validation Results
Dataset 1

Custom Random Forest

vs

Custom Boosted Forest

vs

Sklearn Random Forest

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=4 min_samples_split=2	2.557705	811.718989	2.2042	664.358259	2.767106	877.784507	0.168653	57.736618	2.726358	869.455607
max_depth=4 min_samples_split=4	2.557705	811.718989	2.2042	664.358259	2.767106	877.784507	0.168653	57.736618	2.726358	869.455607
max_depth=4 min_samples_split=8	2.557705	811.718989	2.2042	664.358259	2.767106	877.784507	0.168653	57.736618	2.726358	869.455607
max_depth=8 min_samples_split=8	2.611872	819.214088	2.287481	692.300548	2.861892	898.626918	0.179693	58.637192	2.791565	877.851279
max_depth=8 min_samples_split=4	2.615377	821.211237	2.288626	696.583759	2.877509	901.085383	0.181578	58.657017	2.796955	879.868254
max_depth=8 min_samples_split=2	2.615841	821.691074	2.28988	696.972118	2.880523	901.430047	0.181169	58.733707	2.79701	880.424781
max_depth=16 min_samples_split=8	2.636198	827.581146	2.292341	694.883541	2.890765	906.737227	0.18244	60.49935	2.818638	888.080496
max_depth=16 min_samples_split=4	2.648115	832.773609	2.299541	703.061923	2.923805	912.464389	0.187953	60.843197	2.836068	893.616807
max_depth=16 min_samples_split=2	2.650679	833.716348	2.304286	704.577085	2.928581	913.214002	0.188107	61.404053	2.838786	895.120401

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=4 min_samples_split=2	2.598224	814.974827	2.296714	692.427318	2.811796	882.782371	0.153027	50.406694	2.751251	865.381521
max_depth=4 min_samples_split=4	2.598224	814.974827	2.296714	692.427318	2.811796	882.782371	0.153027	50.406694	2.751251	865.381521
max_depth=4 min_samples_split=8	2.598224	814.974827	2.296714	692.427318	2.811796	882.782371	0.153027	50.406694	2.751251	865.381521
max_depth=8 min_samples_split=8	2.648981	826.11637	2.339354	721.477129	2.987461	910.844044	0.184327	56.167093	2.833307	882.283464
max_depth=8 min_samples_split=2	2.651796	826.933861	2.342117	713.716809	2.991306	912.98982	0.1853	57.449111	2.837096	884.382972
max_depth=8 min_samples_split=4	2.654886	827.924187	2.349587	717.557903	3.019108	921.903541	0.186712	57.41537	2.841598	885.339557
max_depth=16 min_samples_split=8	2.672531	836.409115	2.348063	722.974605	3.006283	918.026752	0.186898	55.790925	2.859428	892.20004
max_depth=16 min_samples_split=4	2.686085	840.95406	2.362696	728.148191	3.012821	922.178043	0.185754	56.494297	2.87184	897.448357
max_depth=16 min_samples_split=2	2.695981	845.966441	2.379354	738.385377	3.036482	930.696452	0.189719	58.439134	2.8857	904.405575

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=4 min_samples_split=2	2.560284	814.575009	2.218556	673.242007	2.7672	882.312868	0.165625	55.518606	2.725909	870.093616
max_depth=4 min_samples_split=4	2.560284	814.575009	2.218556	673.242007	2.7672	882.312868	0.165625	55.518606	2.725909	870.093616
max_depth=4 min_samples_split=8	2.560284	814.575009	2.218556	673.242007	2.7672	882.312868	0.165625	55.518606	2.725909	870.093616
max_depth=8 min_samples_split=8	2.633495	825.143919	2.292886	693.486306	2.967434	919.810773	0.194178	63.874519	2.827673	889.018438
max_depth=8 min_samples_split=2	2.641956	831.962793	2.332882	717.600764	2.991986	926.954148	0.195292	62.020384	2.837248	893.983176
max_depth=8 min_samples_split=4	2.647997	833.480353	2.319163	711.003012	2.972441	929.297937	0.19432	63.429706	2.842317	896.910059
max_depth=16 min_samples_split=8	2.670385	838.538118	2.316965	700.822473	2.989424	926.686447	0.192256	65.428455	2.862641	903.966573
max_depth=16 min_samples_split=4	2.721222	859.777307	2.340236	720.866627	3.055894	952.193865	0.204608	66.873776	2.92583	926.651083
max_depth=16 min_samples_split=2	2.737333	867.118453	2.357501	735.032418	3.075265	969.273146	0.207753	66.991311	2.945086	934.109764

Cross-validation Results

Dataset 2

Custom Regression Tree

vs

Sklearn Regression Tree

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=4 min_samples_split=4	6.422422	59544.266089	3.091338	36711.809393	9.230816	79532.338436	1.618359	12558.233402	8.040781	72102.499491
max_depth=16 min_samples_split=4	6.292019	57896.798977	3.418357	32002.728058	8.741315	72841.641675	1.874404	15130.850262	8.166423	73027.649238
max_depth=8 min_samples_split=4	6.292019	57896.798977	3.418357	32002.728058	8.741315	72841.641675	1.874404	15130.850262	8.166423	73027.649238
max_depth=4 min_samples_split=2	6.499006	60449.208298	3.098786	35725.738306	9.230816	89520.747795	1.702571	14305.449719	8.201577	74754.658018
max_depth=8 min_samples_split=2	6.260765	58879.783340	3.349589	28752.639630	9.222970	83714.011457	1.911328	17531.506116	8.172093	76411.289456
max_depth=16 min_samples_split=2	6.239150	58831.649244	3.379227	28354.893758	9.132840	83489.092530	1.888763	17583.629337	8.127912	76415.278582
max_depth=4 min_samples_split=8	6.265057	57768.741201	3.478629	37282.490418	9.369635	102476.594308	1.772426	18669.939703	8.037483	76438.680903
max_depth=16 min_samples_split=8	6.269276	57409.975182	3.307420	31436.197809	9.947958	104694.474763	2.084322	20597.140532	8.353598	78007.115714
max_depth=8 min_samples_split=8	6.269276	57409.975182	3.307420	31436.197809	9.947958	104694.474763	2.084322	20597.140532	8.353598	78007.115714

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=4 min_samples_split=2	6.167716	56202.418604	3.098786	35725.738306	9.230816	71771.213887	1.503333	11038.035457	7.671049	67240.454061
max_depth=8 min_samples_split=4	5.944121	56119.014581	3.759957	33480.626950	9.072440	72449.654112	1.766872	13590.860060	7.710993	69709.874641
max_depth=4 min_samples_split=4	6.078550	56219.435523	3.091338	36711.809393	8.320996	79532.338436	1.345826	13651.449018	7.424376	69870.884542
max_depth=16 min_samples_split=4	5.998543	56471.796154	3.759957	33480.626950	9.072440	72449.654112	1.841971	13841.607179	7.840515	70313.403333
max_depth=8 min_samples_split=2	6.102438	56529.348501	4.024010	32627.770818	9.144845	81185.853799	1.605755	14983.273306	7.708193	71512.621807
max_depth=16 min_samples_split=2	6.160151	58884.300509	4.053648	34622.453491	8.582648	83704.411899	1.492156	15715.092710	7.652306	74599.393219
max_depth=4 min_samples_split=8	6.360784	58150.706781	4.435898	41102.146223	9.369635	102476.594308	1.624925	18238.328921	7.985709	76389.035702
max_depth=16 min_samples_split=8	6.182865	56924.097094	4.264689	35883.657354	9.947958	104694.474763	1.770224	19765.077878	7.953089	76689.174973
max_depth=8 min_samples_split=8	6.182865	56924.097094	4.264689	35883.657354	9.947958	104694.474763	1.770224	19765.077878	7.953089	76689.174973

Cross-validation Results

Dataset 2

Custom Random Forest

vs

Custom Boosted Forest

vs

Sklearn Random Forest

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=16 min_samples_split=2	7.088698	65882.945442	3.706762	33231.663902	11.560904	91952.362201	2.498170	19443.479582	9.586868	85326.425024
max_depth=16 min_samples_split=4	7.028418	65371.222271	3.413017	31356.543683	11.560344	93273.635151	2.553470	20051.875869	9.581888	85423.098139
max_depth=8 min_samples_split=2	7.106505	65964.761980	3.639740	32998.271244	11.599409	92514.114611	2.507671	19534.239018	9.614176	85499.000998
max_depth=8 min_samples_split=4	7.055574	65490.132929	3.433949	31345.287284	11.581960	93838.852819	2.553335	20118.940935	9.608908	85609.073864
max_depth=4 min_samples_split=4	7.248944	65770.980068	3.243445	29562.199636	12.113330	94569.810255	2.645114	20166.099270	9.894058	85937.079339
max_depth=4 min_samples_split=2	7.290472	66100.414567	3.320804	29947.887865	12.116273	94163.476245	2.627033	19883.788001	9.917505	85984.202568
max_depth=16 min_samples_split=8	7.101694	65089.071863	3.203412	29126.716424	11.808389	97955.271036	2.623151	21340.381594	9.724845	86429.453457

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=16 min_samples_split=8	6.631873	61262.516474	3.264375	29078.824089	10.767887	95733.664562	2.361941	20758.473229	8.993814	82020.989703
max_depth=4 min_samples_split=4	6.997143	62840.223905	3.860007	30867.799619	11.062277	95402.448059	2.234124	19326.319225	9.231268	82166.543130
max_depth=8 min_samples_split=8	7.058529	62696.801153	3.744060	31013.514158	11.608262	92102.609147	2.527323	20213.374619	9.585852	82910.175772
max_depth=16 min_samples_split=2	6.560452	60677.208040	2.930010	24690.266445	9.846970	103145.152923	2.315990	23115.341607	8.876442	83792.549647
max_depth=16 min_samples_split=4	6.658539	62683.519065	3.014915	26543.296920	10.311741	96300.780459	2.332752	21125.183709	8.991291	83808.702774
max_depth=8 min_samples_split=4	6.668917	61356.740383	2.791898	29467.373260	10.782199	102288.283451	2.611279	22561.390698	9.280197	83918.131080
max_depth=4 min_samples_split=8	6.932028	62770.088415	3.239827	29039.304671	11.532225	96695.909571	2.611119	21758.902507	9.543147	84528.990923

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=4 min_samples_split=2	5.625556	50485.653155	3.635754	30911.367365	7.970809	71215.141579	1.662596	12936.306770	7.288152	63421.959925
max_depth=8 min_samples_split=4	5.488087	49589.679244	2.985775	25640.943862	7.433472	77904.922662	1.833896	15789.552364	7.321984	65379.231608
max_depth=16 min_samples_split=4	5.468974	49615.373648	2.917673	25348.340336	7.417884	77829.810415	1.860554	15932.413658	7.329528	65547.787306
max_depth=8 min_samples_split=2	5.404883	49922.537655	3.145510	25799.973385	7.964523	83672.842747	1.615914	16694.223743	7.020797	66616.761398
max_depth=16 min_samples_split=2	5.300504	49210.660386	3.095748	25602.338063	8.086777	87626.084505	1.624764	17839.701981	6.925268	67050.362367
max_depth=4 min_samples_split=4	5.661461	50982.177684	2.761188	28311.860848	7.940817	87610.876195	1.844279	17268.826623	7.505741	68251.004307
max_depth=8 min_samples_split=8	5.919460	53017.256766	3.390409	27989.098102	8.333900	92233.236242	1.759275	17869.674013	7.678735	70886.930779

Cross-validation Results

Dataset 3

Custom Regression Tree

vs

Sklearn Regression Tree

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=24 min_samples_split=16	4.493810	2.686941	4.138510	2.546118	5.245213	2.991959	0.353164	0.152271	4.846974	2.839213
max_depth=12 min_samples_split=16	4.492009	2.685500	4.126555	2.536551	5.254059	2.994621	0.356868	0.155523	4.848877	2.841023
max_depth=12 min_samples_split=8	4.700937	2.804574	4.394020	2.648316	5.391999	3.093829	0.359795	0.157654	5.060733	2.962228
max_depth=24 min_samples_split=8	4.732613	2.832677	4.442100	2.676414	5.424519	3.115121	0.353057	0.152750	5.085670	2.985426
max_depth=12 min_samples_split=4	4.955270	2.957601	4.661447	2.755932	5.773969	3.288618	0.376606	0.168444	5.331877	3.126045
max_depth=24 min_samples_split=4	5.093170	3.046548	4.749620	2.865801	5.940528	3.341763	0.373283	0.154471	5.466452	3.201019
max_depth=6 min_samples_split=16	5.422448	3.192869	4.964107	2.967058	6.382464	3.536621	0.434024	0.188831	5.856471	3.381700
max_depth=6 min_samples_split=4	5.422448	3.192869	4.964107	2.967058	6.382464	3.536621	0.434024	0.188831	5.856471	3.381700
max_depth=6 min_samples_split=8	5.422448	3.192869	4.964107	2.967058	6.382464	3.536621	0.434024	0.188831	5.856471	3.381700

	mean		min		max		std		SCORE	
METRIC	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
PARAMETERS										
max_depth=12 min_samples_split=16	4.487183	2.684070	4.126555	2.535978	5.250976	2.993264	0.356842	0.155219	4.844025	2.839289
max_depth=24 min_samples_split=16	4.495165	2.687855	4.138510	2.546536	5.261918	2.990601	0.358466	0.153155	4.853631	2.841011
max_depth=12 min_samples_split=8	4.699353	2.806547	4.375979	2.644497	5.412876	3.103826	0.370459	0.162465	5.069812	2.969012
max_depth=24 min_samples_split=8	4.733688	2.837109	4.425740	2.674607	5.440028	3.137025	0.364666	0.158657	5.098354	2.995766
max_depth=12 min_samples_split=4	4.963375	2.966024	4.648895	2.771107	5.713662	3.320591	0.358172	0.171651	5.321547	3.137675
max_depth=24 min_samples_split=4	5.098476	3.054856	4.759563	2.871881	5.919104	3.374649	0.370041	0.160288	5.468517	3.215144
max_depth=6 min_samples_split=16	5.422448	3.192869	4.964107	2.967058	6.382464	3.536621	0.434024	0.188831	5.856471	3.381700
max_depth=6 min_samples_split=4	5.422448	3.192869	4.964107	2.967058	6.382464	3.536621	0.434024	0.188831	5.856471	3.381700
max_depth=6 min_samples_split=8	5.422448	3.192869	4.964107	2.967058	6.382464	3.536621	0.434024	0.188831	5.856471	3.381700

Cross-validation Results
Dataset 3

Custom Random Forest

vs

Custom Boosted Forest

vs

Sklearn Random Forest

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=6 min_samples_split=16	10.456630	5.438398	9.368751	5.047590	12.267606	5.766428	0.928469	0.243655	11.385098	5.682052
max_depth=6 min_samples_split=4	10.456642	5.438888	9.368125	5.049918	12.268865	5.766443	0.928973	0.243391	11.385615	5.682279
max_depth=6 min_samples_split=8	10.456731	5.438937	9.368125	5.049905	12.268865	5.766443	0.928941	0.243364	11.385672	5.682301
max_depth=12 min_samples_split=16	10.423309	5.478822	9.323053	4.993917	12.235044	5.935629	0.936266	0.287487	11.359575	5.766308
max_depth=24 min_samples_split=16	10.455144	5.499086	9.368042	5.012603	12.240925	5.964701	0.928939	0.287332	11.384083	5.786418
max_depth=12 min_samples_split=8	10.581569	5.595562	9.451444	5.094807	12.343467	6.068831	0.950560	0.294980	11.532129	5.890542
max_depth=12 min_samples_split=4	10.633034	5.642898	9.477775	5.128770	12.378202	6.121156	0.958285	0.300405	11.591319	5.943302

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=6 min_samples_split=16	8.339887	4.395359	7.658788	4.275143	9.024794	4.594665	0.447233	0.095308	8.787120	4.490667
max_depth=6 min_samples_split=8	8.347603	4.401244	7.657789	4.269229	9.106361	4.618937	0.468923	0.101341	8.816525	4.502585
max_depth=6 min_samples_split=4	8.356842	4.407000	7.657789	4.269229	9.134308	4.621158	0.469131	0.101838	8.825973	4.508838
max_depth=12 min_samples_split=16	8.277321	4.410745	7.688811	4.230152	8.967531	4.591719	0.431488	0.114243	8.708809	4.524988
max_depth=24 min_samples_split=16	8.310836	4.426824	7.763332	4.299686	9.040053	4.601493	0.438430	0.102795	8.749266	4.529619
max_depth=12 min_samples_split=8	8.372736	4.486857	7.864840	4.282493	9.070110	4.679387	0.426890	0.111634	8.799626	4.598492
max_depth=24 min_samples_split=8	8.474242	4.543158	7.955504	4.353714	9.255390	4.799944	0.462188	0.124130	8.936430	4.667288

	mean		min		max		std		SCORE	
METRIC	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**	**MAPE**	**RMSE**
PARAMETERS										
max_depth=12 min_samples_split=16	4.041373	2.379352	3.660776	2.233610	4.978556	2.704586	0.411052	0.176818	4.452424	2.556170
max_depth=24 min_samples_split=16	4.042501	2.378765	3.666755	2.227290	4.975465	2.710508	0.407820	0.177675	4.450322	2.556440
max_depth=12 min_samples_split=8	4.049948	2.399021	3.658579	2.233019	4.915626	2.735651	0.397867	0.175299	4.447816	2.574320
max_depth=24 min_samples_split=8	4.050983	2.400132	3.657965	2.237689	4.927198	2.728040	0.401535	0.175041	4.452517	2.575173
max_depth=12 min_samples_split=4	4.119285	2.442860	3.729191	2.280381	4.947380	2.780280	0.378941	0.170163	4.498226	2.613023
max_depth=24 min_samples_split=4	4.140478	2.455739	3.731825	2.284589	4.947672	2.783120	0.378451	0.170439	4.518929	2.626178
max_depth=6 min_samples_split=4	4.635746	2.704301	4.147550	2.521260	5.671995	3.034688	0.473999	0.201464	5.109745	2.905765